

Global Supply Chain & Logistics Performance

Final Project Report

Student Information

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Track: Operations Analytics

Project Title: Global Supply Chain & Logistics Performance

1. Introduction

The project focuses on analyzing supply chain operations using the DataCo Supply Chain Dataset. The objective is to understand logistics activities, shipping performance, sales trends, customer behavior, and operational efficiency through data analytics.

2. Objectives

- Understand the structure of the supply chain dataset.
- Analyze logistics and shipping operations.
- Identify sales and profit trends.
- Examine delivery performance.
- Evaluate customer and market behavior.
- Generate business insights for supply chain optimization.

3. Tools and Technologies Used

Google Colab, Python, Pandas, NumPy, Matplotlib, Seaborn, GitHub.

4. Dataset Description

Dataset Used: DataCo Supply Chain Dataset. The dataset contains information related to orders, customers, products, sales, profits, shipping modes, delivery status, markets, countries, and regions.

5. Project Workflow

Notebook 01: Data Loading – Imported libraries, uploaded dataset, extracted ZIP files, loaded CSV dataset, verified structure, and created backup.

Notebook 02: Data Understanding – Explored features, data types, summary statistics, missing values, and visualizations.

Notebook 03: Data Cleaning – Handled missing values, removed duplicates, treated outliers, and saved cleaned data.

Notebook 04: EDA – Analyzed sales, profits, shipping modes, delivery status, regions, markets, and correlations.

6. Key Findings

- Sales were concentrated in specific categories and regions.
- Profitability varied across products and regions.
- Certain shipping modes were used more frequently.
- Customer purchasing behavior showed identifiable trends.
- Multiple international markets contributed to growth.

7. Business Insights

Improving delivery efficiency can enhance customer satisfaction. High-performing regions deserve additional focus, while underperforming markets may require operational improvements.

Data-driven inventory planning can reduce logistics costs.

8. Challenges Faced

Handling missing values, cleaning records, detecting outliers, and understanding logistics variables.

9. Conclusion

The project successfully analyzed supply chain operations using the DataCo Supply Chain Dataset

and generated valuable insights regarding sales performance, customer behavior, shipping operations, and market trends.

10. Future Scope

Predict shipment delays, forecast sales, build dashboards, optimize inventory, and implement advanced supply chain analytics solutions.

References

DataCo Supply Chain Dataset, Python Documentation, Pandas, NumPy, Matplotlib, Seaborn, and Google Colab Documentation.

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